

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 07



By- SATISH YADAV SIR

TOPICS TO BE COVERED

01 Basics of coprime

02 Euler totient functions

03 Problems in Totient Function

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coprime./Relative prime. :

$\text{gcd}(a, b) = 1$. a, b are relative prime/coprime.

$\text{gcd}(3, 5) = 1$. $\text{gcd}(3, 6) \neq 1$

$\text{gcd}(3, 7) = 1$.

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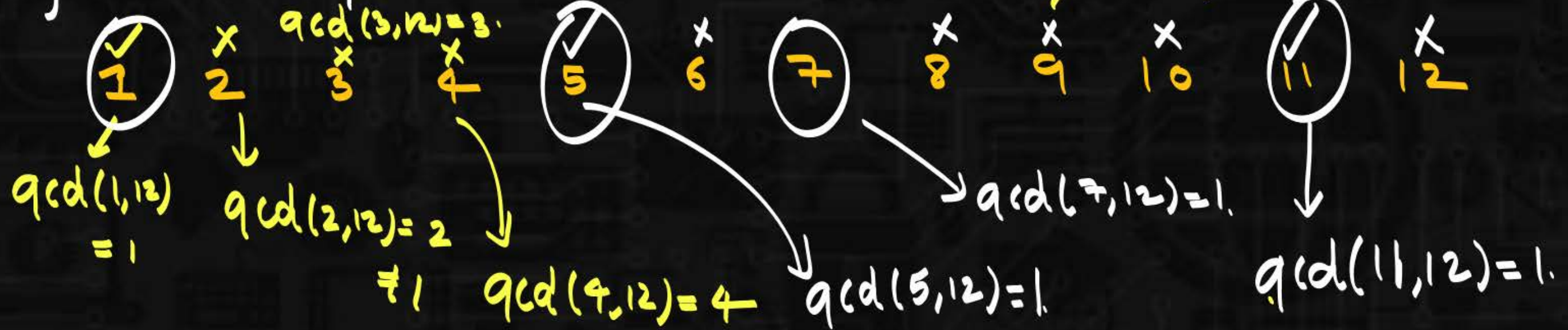


$\phi(n)$: no. of relative primes less than n .

12

no. of relative primes are less than 12. = $\phi(12)$

$$\gcd(12, 12) = 12$$

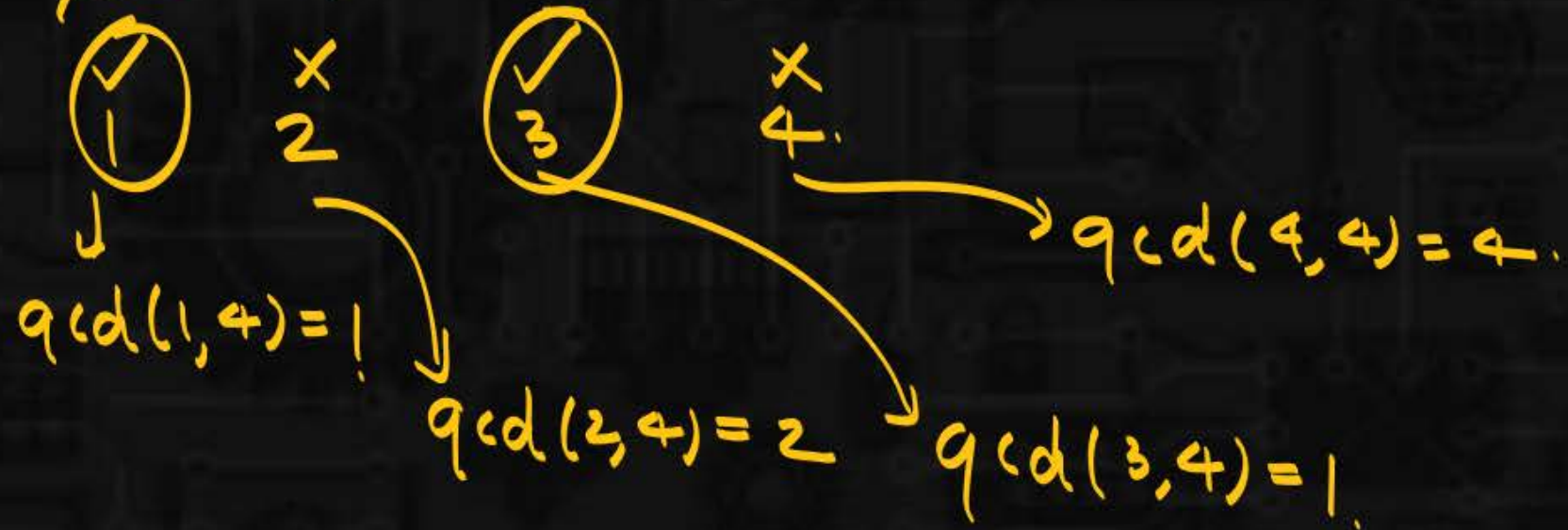


$$\phi(12) = 4$$

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$$\phi(4) = 2.$$



$$\phi(3) = 2$$



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$$\frac{\phi(12) = 4}{\downarrow} \quad \phi(3) = 2$$

$$\phi(4) = 2$$

$$\phi(12)$$

$$\downarrow$$

$$\phi(3 \times 4)$$

$$\gcd(3, 4) = 1$$

$$\frac{\phi(3)}{\downarrow 2} \times \frac{\phi(4)}{\downarrow 2} = 4$$

ϕ multiplicative in nature

$$\phi(A \cdot B) = \phi(A) \cdot \phi(B)$$

$$\gcd(A, B) = 1$$

A, B should be relative prime

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$$\phi(7) = 6$$

① ② ③ 4 5 6 ⑦^x

$$\gcd(2, 7) = 1$$

$$\gcd(3, 7) = 1$$

$$\gcd(4, 7) = 1$$

$$\gcd(5, 7) = 1$$

$$\gcd(6, 7) = 1$$

$$\phi(p) = p-1$$

p is prime.

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Relative prime

$$\phi(49) = \text{Total} - \text{non Relative prime.} = 49 - \frac{49}{7}$$



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Relative prime. = Total - non Relative prime.

$$\phi(49) = 49 - \frac{49}{7}$$

$$\phi(7^2) = 7^2 - \frac{7^2}{7}$$

$$\phi(p^a) = p^a - \frac{p^a}{p} \quad a \geq 1$$

$$= p^a \left[1 - \frac{1}{p} \right]$$

$$= p^a \left[\frac{p-1}{p} \right]$$

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$$n = p_1^a \cdot p_2^b \cdot p_3^c \dots \left[\begin{array}{l} p_1, p_2, p_3, \dots \\ \text{prime no.} \end{array} \quad a, b, c \geq 0 \right]$$

$$\phi(n) = \phi(p_1^a \cdot p_2^b \cdot p_3^c \dots)$$

$$= \phi(p_1^a) \times \phi(p_2^b) \times \phi(p_3^c) \dots$$

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$$\phi(n) = \phi(p_1^a) \times \phi(p_2^b) \times \phi(p_3^c) \dots$$

$$= \left(p_1^a - \frac{p_1^a}{p_1} \right) \times \left(p_2^b - \frac{p_2^b}{p_2} \right) \times \left(p_3^c - \frac{p_3^c}{p_3} \right)$$

$$= p_1^a \left[1 - \frac{1}{p_1} \right] \times p_2^b \left[1 - \frac{1}{p_2} \right] \times p_3^c \left[1 - \frac{1}{p_3} \right] \dots$$

$$\phi(n) = p_1^a \cdot p_2^b \cdot p_3^c \frac{[p_1 - 1][p_2 - 1][p_3 - 1] \dots}{p_1 \cdot p_2 \cdot p_3 \dots}$$

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$$\phi(n) = \frac{p_1^a \cdot p_2^b \cdot p_3^c}{p_1 \cdot p_2 \cdot p_3} [p_1 - 1][p_2 - 1][p_3 - 1]$$

$$n = p_1^a \cdot p_2^b \cdot p_3^c$$

$$\phi(n) = n \times \frac{[p_1 - 1][p_2 - 1][p_3 - 1] \dots}{p_1 \cdot p_2 \cdot p_3 \dots}$$

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$$\phi(110) = \frac{n [p_1-1] [p_2-1] [p_3-1]}{p_1 \cdot p_2 \cdot p_3} = \frac{\cancel{110} (2-1)(5-1)(11-1)}{\cancel{2 \cdot 5 \cdot 11}}$$

$$110 = 11 \times 10$$

$$= 11 \times 2 \times 5$$

$$= 1 \cdot 4 \cdot 10 = \underline{\underline{40}}$$

$$\rightarrow 210 \quad \text{Ans: 48}$$

$$\rightarrow 1368 \rightarrow \text{Ans: 432.}$$

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(GATE-05) $n = p^2 q$

p, q are distinct prime

how many numbers m satisfy $1 \leq m \leq n$ $\gcd(m, n) = 1$.

a) $p(q-1)$ c) $(p^2-1)(q-1)$

b) $p \cdot q$

☒ d) $p(p-1)(q-1)$

$$\begin{aligned} \phi(n) &= n \frac{(p_1-1)(p_2-1)}{p_1 \cdot p_2} = \cancel{p^2} \cancel{q} \frac{(p-1)(q-1)}{\cancel{p} \cdot \cancel{q}} \\ &= p(p-1)(q-1) \end{aligned}$$

